Journal of Plant Growth Regulation (2025) 44:155–167
https://doi.org/10.1007/s00344-024-11468-8

Genome-Wide Identification, Evolution, and Characterization of *GATA* Gene Family and *GATA* Gene Expression Analysis Post-MeJA Treatment in *Platycodon grandiflorum*

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Received: 6 April 2024 / Accepted: 19 August 2024 / Published online: 1 September 2024

Abstract
The GATA-binding factor (GATA) plays a major role in regulating plant development and response to distinct environmental stresses. At present, GATAs are characterized in various model plant species, including *Arabidopsis thaliana* and *Oryza sativa*. However, the *GATA* gene family in *Platycodon grandiflorum* is not yet fully understood. The study aimed to develop a comprehensive understanding of the GATA TFs and explore the regulatory mechanism of methyl jasmonate (MeJA) on the GATA members in *P. grandiflorum*. A total of 22 *PgGATAs* were identified based on publicly available genome data of *P. grandiflorum*, and each member was analyzed in detail. The 22 identified genes were distributed across nine chromosomes. Their phylogenetic tree and domain structures showed that the GATAs could be clustered into four subfamilies (A–D). The structural protein domains and conserved motifs of the PgGATA family members were relatively conserved across different subfamilies. Light and hormone response elements were found in abundance in the promoter sequences. In addition, quantitative real-time polymerase chain reaction (qRT-PCR) indicated that *PgGATA4*, *6*, *7*, *8*, and *11* were sensitive to MeJA treatment in *P. grandiflorum* roots. Nevertheless, co-expression network analysis revealed that the activities of the genes in the family remained significantly correlated, suggesting possible synergy in their functions. Two (*PgGATA5* and *PgGATA9*) and three (*PgGATA8*, *PgGATA11*, and *PgGATA22*) hub *PgGATAs* were identified that might have central functions in *P. grandiflorum* tissues and MeJA-treated roots, respectively. This study provided detailed information about the *PgGATA* gene family and facilitated a functional characterization of the candidate genes.

Keywords *Platycodon grandiflorum* · Genome-wide · GATA transcription factors · Gene expression · MeJA

Introduction

GATA transcription factors (TFs) widely exist in animals, plants, and microbes. These factors contain conserved group IV zinc finger structure ($CX_2CX_{17\text{-}20}CX_2C$) and a basal specific region that can bind to WGATAR to regulate the expression of the relevant downstream genes at the transcriptional level (Lowry and Atchley 2000). More members of the GATA family of TFs have been identified in plants than in animals and microorganisms (Romano and Miccio 2020; Scazzocchio 2000). Most GATA TFs contain a single $CX_2CX_{18}CX_2C$ motif, and several TFs in plants contain $CX_2CX_{20}CX_2C$. NTL1 was the first GATA TF identified in tobacco and contains the $CX_2CX_{18}CX_2C$ motif (Daniel-Vedele and Caboche 1993). It was subsequently identified in *Arabidopsis thaliana* (Rayko et al. 2010) and *Oryza sativa* (Gupta et al. 2017).

Numerous studies have confirmed that GATA TFs regulate several genes involved in plant growth and development, light regulation, nitrogen metabolism, and hormone signaling. *AtGATA2* acts as a vital regulator of light morphogenesis by interacting with brassinosteroid signaling in *A. thaliana* (Luo et al. 2010). Studies have demonstrated

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that some *VvGATAs* display differential responses to light and phytohormones, demonstrating their importance in the regulation of light and hormone signaling pathways (Zhang et al. 2018). *GmGATA44* and *GmGATA58* are involved in the nitrogen metabolism in soybean and have been shown to regulate chlorophyll biosynthesis in *A. thaliana* (Zhang et al. 2015a). In terms of secondary metabolite synthesis, *CrGATA1*, a GATA TF with a Leu-Leu-Met domain, is involved in the light-induced regulatory pathway of vindoline biosynthesis in *Catharanthus roseus* seedlings (Liu et al. 2019). In terms of abiotic stress, *SlGATA17* overexpression lines are more drought tolerant than the wild type, indicating that *SlGATA17* alters the drought tolerance of tomato plants by regulating the key enzyme-encoding genes related to the phenylpropanoid biosynthesis pathway (Zhao et al. 2021). In addition, transgenic *A. thaliana* overexpressing *BdGATA13* has been show to exhibit darker the leaf color, delayed flowering time, and enhanced drought resistance (Guo et al. 2021).

Platycodon grandiflorum is a traditional East Asian herbal medicine. It is used to treat cough, tonsillitis, and sore throat, among other respiratory diseases (Zhang et al. 2015b). It is also widely used in health food and vegetable dishes. The pharmaceutical and economic value of this plant has attracted a lot of attention toward the biosynthesis and regulation of bioactive ingredients in *P. grandiflorum*. In addition, methyl jasmonate (MeJA) has been shown to regulate GATA TFs, thereby affecting root growth and development and secondary metabolite biosynthesis. However, there is a lack of an understanding of the molecular regulatory mechanisms underlying non-biotic stress in *P. grandiflorum*, especially with respect to the functions of the *GATA* genes under MeJA stress. The goal of this study was to identify the *P. grandiflorum* GATA family members and comprehensively characterize the genes, conserved motifs, and phylogenetic relationships. Additionally, transcriptome data were used to analyze their expression patterns in multiple tissues, and quantitative real-time polymerase chain reaction (qRT-PCR) was used to analyze the expression patterns of each member of the gene family under MeJA stress.

Materials and Methods

Identification of GATA Family Genes

The *P. grandiflorum* genome was obtained from Figshare (https://doi.org/https://doi.org/10.6084/m9.figshare.21511020). The transcriptome data of different tissue parts (stem, stamen, sepal, seed, root, pistil, petal, and leaf) of *P. grandiflorum* and MeJA treatment were obtained from previous studies (Kim et al. 2020). Transcriptome gene expression levels were quantified using fragments per kilobase million (FPKM) as the unit. First, the known GATA protein sequences of *A. thaliana* were collected from the TAIR database (https://www.arabidopsis.org/) and used for BLASTP searching. Then, the presence of the GATA domain in the identified protein sequences was confirmed using the SMART database (Version 8.0) (http://smart.embl-heidelberg.de/), NCBI-Conserved Domain database (Version 3.21) (https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi), and Pfam (http://pfam.xfam.org/search#tabview=tab1), with an e-value cut-off of 0.01.

The biological information of the GATA proteins was predicted using the ExPASy database (https://web.expasy.org/protparam/), such as the coding sequence (CDS) length, isoelectric point (PI), molecular weight (MW), and instability index. The subcellular locations of the GATA proteins were analyzed using the WoLF PSORT (https://wolfpsort.hgc.jp/).

Chromosomal Locations and Synteny Analysis

The positions of the identified genes on the chromosomes were determined based on the gene numbers in *P. grandiflorum* general feature format files and genome annotation files. Then, all *PgGATAs* were mapped on the chromosomes using TBtools (Chen et al. 2020). The replication pattern for each *PgGATAs* was analyzed via the Multiple Collinearity Scan toolkit (MCscanX) software with default parameters. We randomly selected five species (*Solanum lycopersicum*, *Cucumis sativus*, *Malus pumila*, *Vitis vinifera*, and *Oryza sativa*) for collinearity relationship analysis with *P. grandiflorum*, and the results were exhibited using TBtools.

Phylogenetic Evolution and Classification of the *PgGATA* Gene Family

The amino acid sequences of PgGATA and AtGATA were summarized, and Clustal X 2.0 was used for multiple sequence alignment. Multiple alignment and construction of a maximum likelihood (ML) tree were performed using MEGA 7.0. The parameters included Jones-Taylor-Thornton (JTT), gamma distribution rate, and partial deletion. The bootstrap method was used to test the tree with 1000 replicates. Evolview is used to visualize and beautify phylogenetic trees (Subramanian et al. 2019).

Conserved Motif, Gene Structure, and Promoter Region Analyses

The conserved motifs of the *PgGATAs* were analyzed using the MEME software (https://meme-suite.org/meme/) (Bailey et al. 2009). The maximum number of motifs was 10, and the distribution of motif occurrences was zero or one per sequence. Promoter sequences of a length of around

2000 bp and located upstream from the translation start site of PgGATAs were extracted using TBtools and analyzed by the PlantCARE database (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/) for the analysis and screening of cis-regulatory elements related to plant growth and development, phytohormones, and abiotic stress.

Treatment

Biennial *P. grandiflorum* was grown in a greenhouse in the institute laboratory at 24–28 °C and treated with 100 µm/L MeJA. The roots were collected at 0, 6, 12, 24, and 48 h post-treatment. Three replicates were wrapped in foil paper, frozen in liquid nitrogen, and promptly preserved at – 80 °C for RNA extraction. Total RNA from the roots of different treatment groups was extracted using a plant RNA kit (SIMGEN, Hangzhou, China). cDNA was synthesized using a BlazeTaq™ First-Strand cDNA Synthesis Kit (GeneCopoeia, Rockville, MD, USA). qPCR was performed using BlazeTaq™ SYBR Green qPCR Mix 2.0 (GeneCopoeia, Rockville, MD, USA). Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was chosen as the reference gene (Ma et al. 2016). The primers used in qRT-PCR are listed in Supplementary Table S1. The expression levels of *PgGATAs* were computed using the $2^{-\Delta\Delta CT}$ method. Raw data were analyzed using a one-way analysis of variance and visualized using GraphPad Prism 8.0.

Subcellular Localization of *PgGATA* Genes

Based on genomic information, the full-length sequences of *PgGATA1* and *PgGATA7* were cloned by PCR, and the cDNA of *P. grandiflorum* root was used as the template. The genes and pCambia1300-35S-EGFP vector with the enhanced gene sequence of green fluorescent protein (EGFP) were digested using restriction enzyme and ligated with T4 ligase. All primers are shown in Supplementary Table S2. Then, the pCambia1300-PgGATAs-35S-EGFP and the control pCambia1300-35S-EGFP vectors were transformed into *Agrobacterium tumefaciens* strain GV3101. When the bacterial suspension reached an OD_{600} of 1, it was injected into the leaves of 4- to 6-week-old tobacco plants using needleless syringes. Forty-eight hours post-infiltration, the EGFP signal in the treated tobacco leaves was captured using a fluorescence microscope (Zeiss LCM-800, Oberkochen, Germany).

Protein–Protein Interaction Network of PgGATAs

STRING software (https://cn.string-db.org/) was used to analyze and construct the functional regulatory network of the PgGATAs based on the orthologues in *Arabidopsis* (Szklarczyk et al. 2021). The threshold confidence value was set at 0.8. The confidence scores are an important indicator to evaluate the evidence of protein–protein association, and they range between zero and one (Szklarczyk et al. 2023).

Co-expression Networks of the *PgGATA* Gene Transcripts and the Potential Hub *PgGATA* Genes

To construct the coexpression network of the *PgGATA* gene family, the Pearson correlation coefficients of *PgGATA* gene transcripts expressed in different tissues (stem, stamen, sepal, seed, root, pistil, petal, and leaf) of a two-year-old *P. grandiflorum* plant and roots treated with MeJA for different periods were calculated. The networks were visualized using the R language. To reveal the characteristics of the network, the co-expression networks of the *PgGATAs* were constructed with p values < 0.05 and correlation > 0.7.

Results

Genome-Wide Identification of the *GATA* Gene Family in *P. grandiflorum*

A total of 22 *PgGATAs* were identified in the *P. grandiflorum* genome (Supplementary Table S3). PgGATAs encode polypeptides made up of anywhere from 110 to 790 amino acids, with molecular weights varying between 12.46 kDa to 88.45 kDa. The PI of the 22 PgGATAs ranged from 4.76 to 9.96. Interestingly, the PI of most PgGATAs (13/22) was greater than 7, indicating that PgGATAs might primarily be composed of basic amino acids. In silico subcellular location prediction revealed that GATA members were localized to the nucleus, except for PgGATA8 and PgGATA16, which were localized to the chloroplast and the vacuole, respectively.

Genetic Mapping of Chromosomes, Gene Replication Events, and Collinearity Analysis

Figure 1 shows the chromosomal locations of 22 *PgGATAs* across nine chromosomes. These genes were named according to their chromosomal location. Most genes were located on chromosomes 2, 5, and 6. Furthermore, each of the chromosomes 3, 4, and 9 harbored only one *PgGATA*, while the remaining chromosomes had two to three genes.

In addition, a duplication event was observed on chromosome 2, namely *PgGATA5* and *PgGATA6*. Gene duplication analysis led to the identification of five repeat gene pairs of genes with fragment repetition: *PgGATA4*/*PgGATA19*, *PgGATA11*/*PgGATA13*, *PgGATA11*/*PgGATA17*, *PgGATA13*/*PgGATA17*, and *PgGATA16*/*PgGATA20* (Fig. 2a). Two pairs of gene copies belong to the A group, and three pairs belong to the B group. These replication events can occur

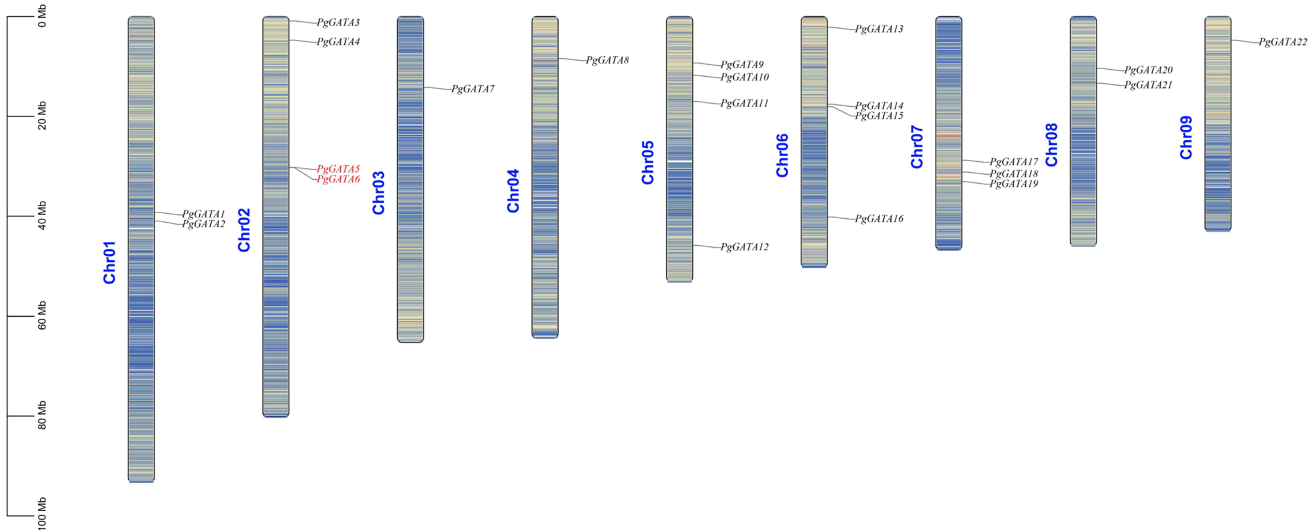


Fig. 1 Chromosome mapping of *GATA* genes in *P. grandiflorum*. Gene names in red indicate duplication events. The chromosome number is displayed on the left side of each chromosome, while the scale on the left indicates the length of the chromosomes

throughout genome replication and were the main drivers of the evolution of *PgGATAs*.

The 22 *PgGATAs* in *P. grandiflorum* contain 35, 23, 42, 30, and 4 collinearity, respectively, with tomato, cucumber, apple, grape, and rice (Fig. 2b). Thus, *P. grandiflorum* shared the most the fewest collinear genes with apple and rice, respectively. The collinearity results showed that the collinear genes might perform analogous functions. Moreover, the collinearity of genes between dicotyledonous plants was significantly higher than that between dicotyledonous and monocotyledonous plants.

Evolutionary Exploration of *PgGATA* Family Genes

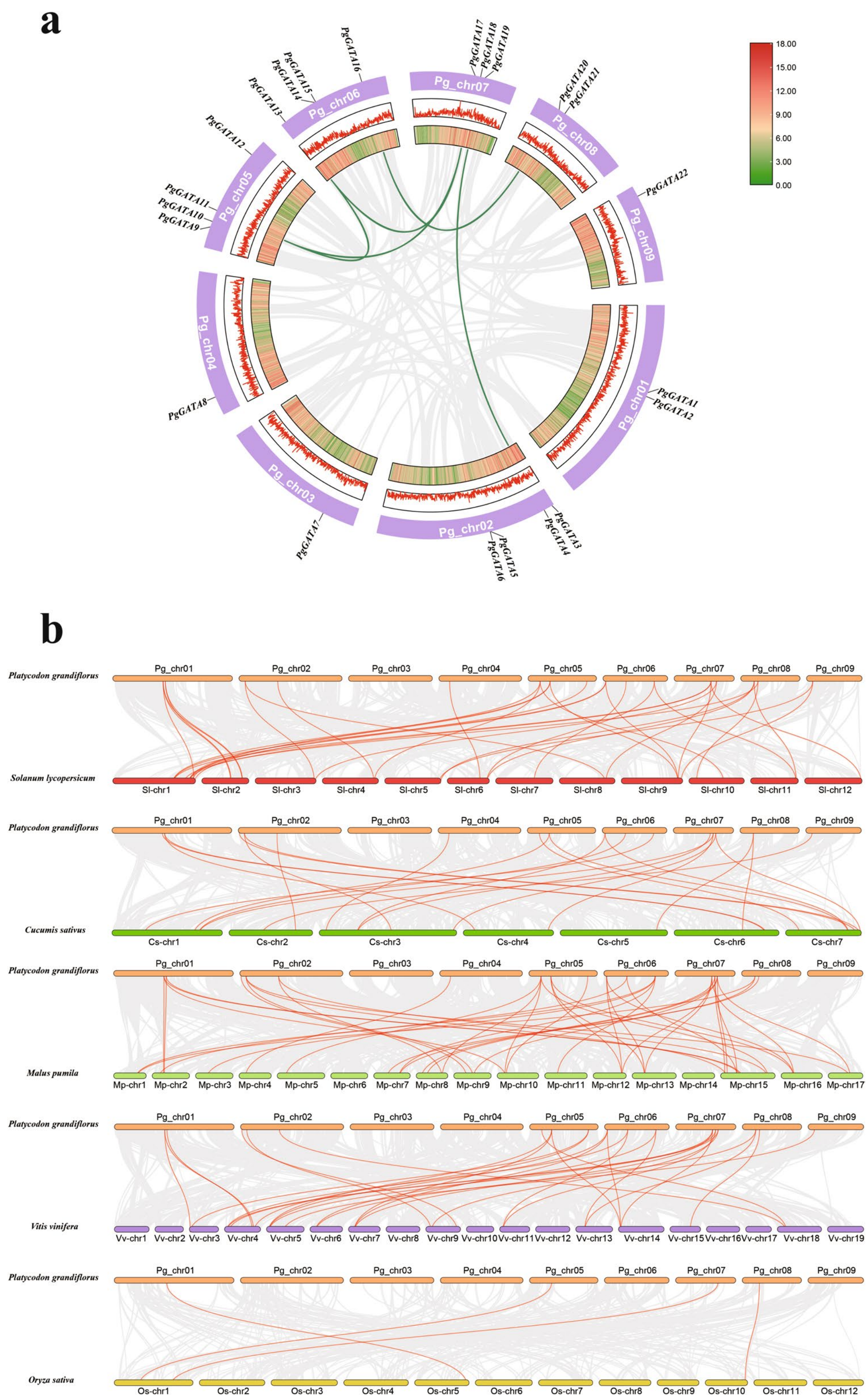
A phylogenetic tree was constructed using the 22 sequences to further analyze the selected sequences (Fig. 3). Additionally, to better highlight the differences between *P. grandiflorum*, *A. thaliana* were included in the evolutionary analysis. The results showed that the PgGATAs were classified into four groups, similar to *A. thaliana* GATAs. Among them, the A group comprised the most genes ($n=9$), followed by the B group ($n=6$), the C group ($n=5$), and the D group (two PgGATAs: PgGATA8 and PgGATA10). Finally, we detected three pairs of direct homologous genes (PgGATA18 and AT4G26150, PgGATA20 and AT3G54810, and PgGATA22 and AT3G24050), indicating that their evolutionary trend is highly consistent.

Analysis of *PgGATA* Gene Structure and Conserved Motifs

The structures and motifs in the 22 *PgGATAs* are shown in Fig. 4. The phylogenetic relationship and classification of GATAs were further supported by motif analysis (Fig. 4a and Supplementary Fig. 1). Group A genes carried the same motifs, and motifs 1, 2, 3, 6, and 7 were found in most members of this group. In group B, most genes contained motifs 1 and 4, PgGATA2 contained motif 1, and PgGATA15 contained motif 2. Most members of group C (PgGATA3, PgGATA5, and PgGATA9) contained motifs 1, 4, 5, and 9, arranged as motif 5 + motif 9 + motif 4 + motif 1. Group D members contained motifs 1 and 6.

At the same time, all members included a GATA domain. Members of groups A and B members contained the CX2CX18CX2C domain, and the CX2CX20CX2C domain was found in group C. However, PgGATA14 lacked the first CX2C structure (Supplementary Fig. 2). In addition, group C contained the CCT and TIFY domains. Group D harbored the PspC and ASXH domains (Fig. 4b). The analysis of introns and exons in *PgGATAs* (Fig. 4c) showed that the members of groups C and D harbored more CDSs than the members of groups A and B.

Fig. 2 Collinearity analysis of *PgGATAs*. **a** Gene duplication of the *GATA* genes of *P. grandiflorum.* **b** Multiple collinearity analysis of *GATA* genes between *P. grandiflorum* and five plant species



Prediction of *Cis*-Acting Elements of *PgGATAs*

To further elucidate the mechanism underlying the activities of *PgGATAs*, the composition and distribution of the cis-acting elements in the promoter sequence of each family member were extracted from the PlantCare database and investigated (Fig. 5). In total, 41 types of cis-acting elements were detected and grouped into three broader

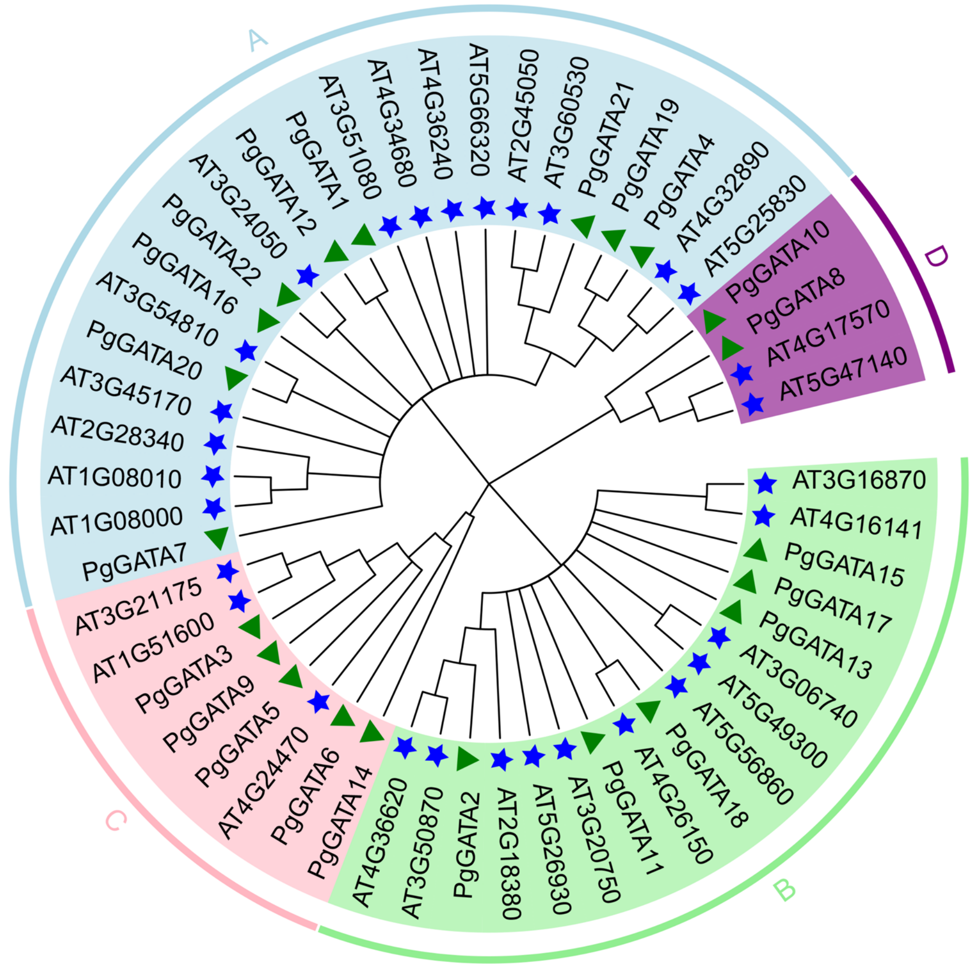


Fig. 3 Phylogenetic tree of the GATA proteins from *Arabidopsis* and *P. grandiflorum*. Different colors represent different subfamilies as follows: Subfamilies A (light blue), B (light green), C (pink), and D (purple). The blue pentagram and green triangle represent the members of the *Arabidopsis* and *P. grandiflorum* gene families, respectively

categories: Phytohormone ($n = 133$), and abiotic and biotic stress ($n = 76$), plant growth and development ($n = 299$). Box 4 cis-acting elements were distributed across all members except PgGATA6. In addition, most genes had the two light-responsive cis-acting elements: G-box and GT1 motifs.

Analysis of *GATA* Gene Expression in Different Tissue Parts of *P. grandiflorum* and Expression of *PgGATA*s Under MeJA Stress

We analyzed the expression levels of *PgGATA* gene family members in different tissues (stem, stamen, sepal, seed, root, pistil, petal, and leaf) based on the transcriptome data from previous studies (Fig. 6a) (Kim et al. 2020). In groups A, B, and D, only *PgGATA1*, *PgGATA20*, *PgGATA15*, and *PgGATA10* showed tissue specificity. Among group C members, *PgGATA5*, *PgGATA6*, *PgGATA9*, and *PgGATA14* were highly expressed in stamens, while *PgGATA3* was highly expressed in the leaves and stems. These results indicated that members of the PgGATA family were primarily involved in stamen growth and development.

MeJA is generally used as an important hormone for abiotic stress treatment and regulates secondary metabolites in medicinal plants. Previous transcriptome data of MeJA-treated *P. grandiflorum* showed that five genes (*PgGATA4*, *PgGATA6*, *PgGATA7*, *PgGATA8*, and *PgGATA11*) were highly expressed at 48 h post-MeJA treatment, while other genes showed low expression (Fig. 6b) (Kim et al. 2020). To further verify the accuracy of the transcriptome data and explore the effects of MeJA on the expressions of *PgGATA4*, *PgGATA6*, *PgGATA7*, *PgGATA8*, and *PgGATA11* in *P. grandiflorum* roots, we treated two-year-old *P. grandiflorum* with MeJA for varying time periods (0, 6, 12, 24, and 48 h) before subjecting them to qPCR (Fig. 7). Our results indicated that the relative expression levels of *PgGATA4* and *PgGATA6* were the highest at 12 h post-MeJA treatment, while the expressions other three genes (*PgGATA7*, *PgGATA8*, and *PgGATA11*) peaked at 48 h after MeJA treatment.

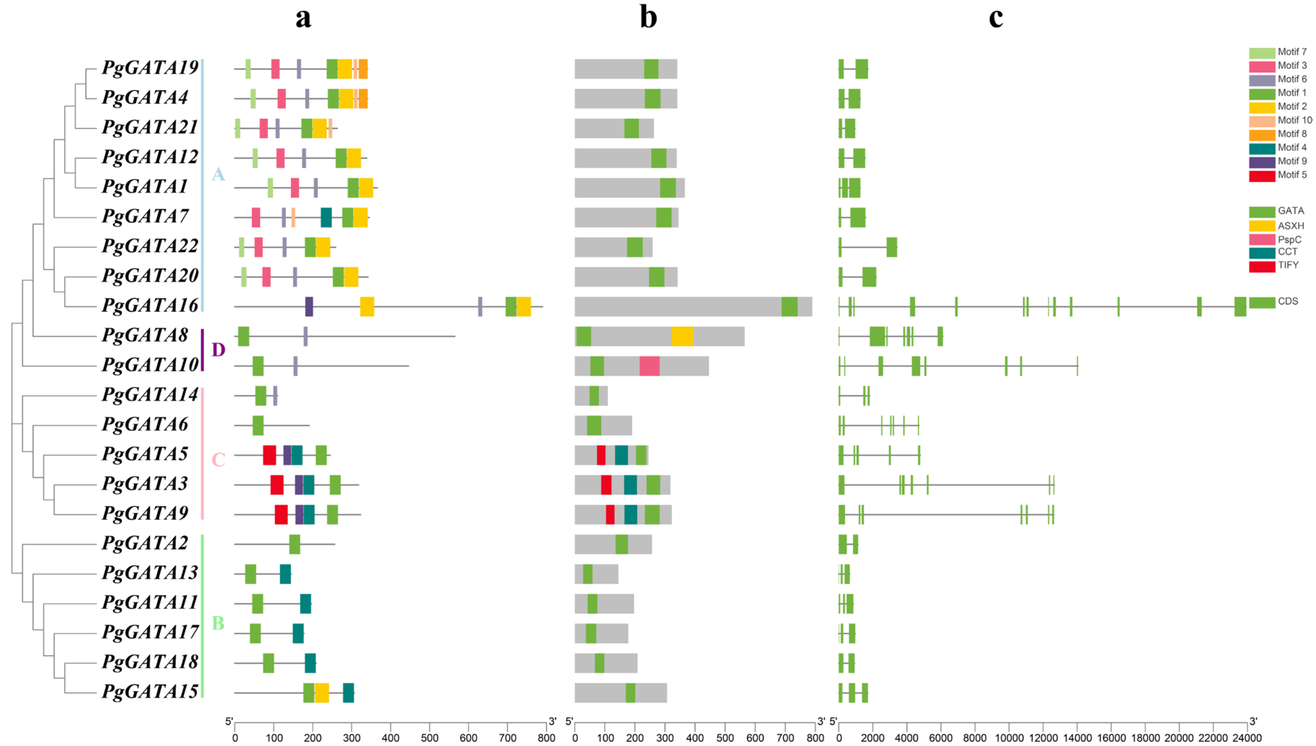


Fig. 4 Phylogenetic relationships, motif distributions, domain distributions, and exon–intron structures of *P. grandiflorum GATA* genes. **a** The amino acid motifs in PgGATA are shown in boxes with different colors. **b** The amino acid domains in PgGATA are shown in colored boxes. **c** Green rectangles represent exons, and gray lines represent introns

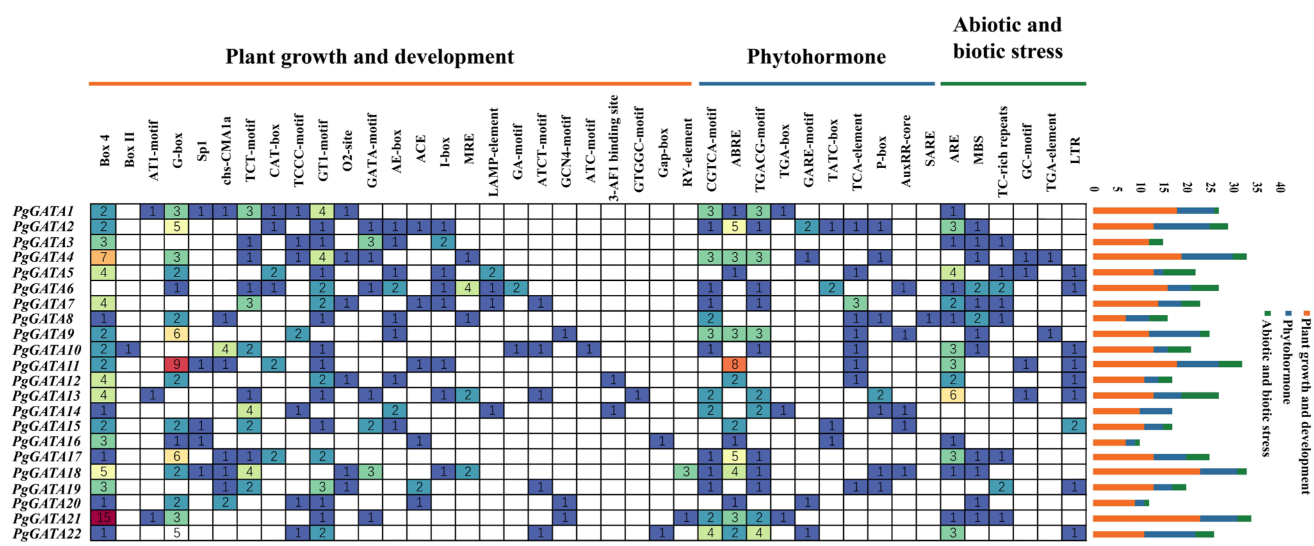


Fig. 5 Cis-acting components of *PgGATA* genes

Protein–Protein Interaction Network Analysis of PgGATA Family Members

The STRING software was used for predictive analysis to further explore the interactions among *GATA* family members at the protein level (Fig. 8). Eighteen proteins formed protein–protein interactions. GATA4 (*PgGATA4* and *PgGATA7*) only interacted with GATA26 (*PgGATA8* and *PgGATA10*); GATA15 (*PgGATA11* and *PgGATA17*), GATA1 (*PgGATA15* and *PgGATA22*), GATA17

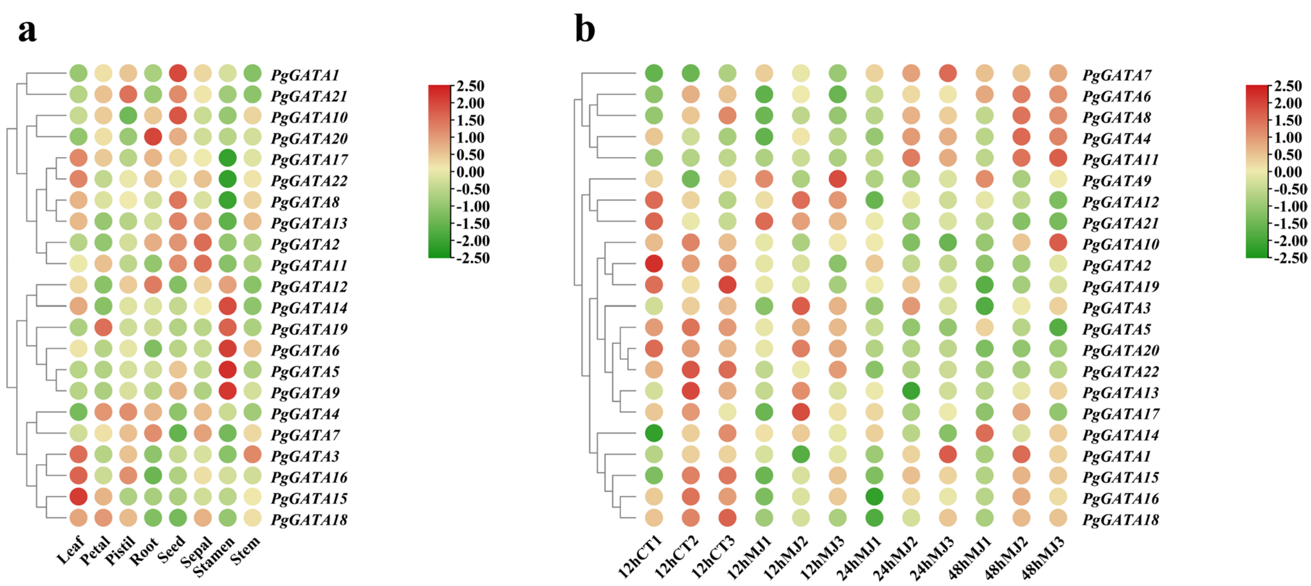


Fig. 6 Heatmap showing the expression profiles of the 22 *PgGATAs* in different tissue sites and under MeJA treatment over different time points. **a** Expression patterns *PgGATAs* between *P. grandiflorum* leaf, petal, pistil, root, seed, sepal, stamen, and stem. **b** Expression patterns of *PgGATAs* in response to MeJA treatment over different time points. MJ 12 h, MJ 24 h, and MJ 48 h represent plants treated with MeJA for 12, 24, and 48 h, respectively; 12 h CT indicates the control

(*PgGATA13*), and GATA19 (*PgGATA2*) exhibited two interaction relationships with other proteins; BME3 (*PgGATA20*) and ZIM (*PgGATA5*) exhibited three interaction relationships with other proteins; and GATA26 (*PgGATA8* and *PgGATA10*) and ZML2 (*PgGATA3*) exhibited four interaction relationships with other proteins. ZML1 (*PgGATA9*) showed seven interactions with other proteins.

Co-expression Networks of the *PgGATAs* Based on Transcriptome

To estimate the functional relationships within the *PgGATA* gene family, their co-expression networks were constructed using their expression profiles in different tissues and in MeJA-treated *P. grandiflorum* roots over different time points. Network construction was carried out under the screening conditions of a P value > 0.05 and correlation > 0.7 to identify the hub genes that play a pivotal role in the network, especially those that interacted with the largest number of other genes in the network. Our results showed that 16 members appeared in the co-expression network of different tissues, with a total of 15 edges, where *PgGATA5* and *PgGATA9* were emphasized as potential hub genes at the core of the network (Fig. 9a). In addition, an 18 members and 38 correlations, including 31 positive correlations and seven negative correlations, appeared in the co-expression network of MeJA-induced genes, where *PgGATA8*, *PgGATA11*, and *PgGATA*22 were the potential hub genes in the network (Fig. 9b).

Subcellular Localization of *PgGATAs*

The full-length CDSs of *PgGATA1* and *PgGATA7* were fused with EGFP to construct the pCambia1300-35S-PgGATAs-EGFP vectors. The pCambia1300-35S-EGFP vector was used as the positive control. All vectors were transiently expressed in tobacco leaf. The EGFP signal of the positive control was detected around the cytomembrane and the nucleus. The fluorescence signal of PgGATAs-EGFP was predominantly detected in the nucleus of tobacco leaf epidermal cells, which was consistent with the previous predication (Fig. 10).

Discussion

The GATA family has been extensively characterized in several plant species, such as *Brachypodium distachyon* (28) (Peng et al. 2021), wheat (79) (Feng et al. 2022), *Brassica napus* (96) (Zhu et al. 2020), *Cucumis sativus* (26) (Zhang et al. 2021), and *Ophiorrhiza pumila* (18) (Shi et al. 2022). However, the composition of the GATAs in *P. grandiflorum* are still not fully understood. In the current study, in the A group, the PIs of PgGATA7, PgGATA16, PgGATA21, and PgGATA22 were greater than 7, and those of PgGATA1, PgGATA4, PgGATA12, PgGATA19, and PgGATA20 were less than 7. The PIs of all the members of groups B and D were greater than 7. Thus, all the members of these two groups were alkaline. In contrast, all the members of group

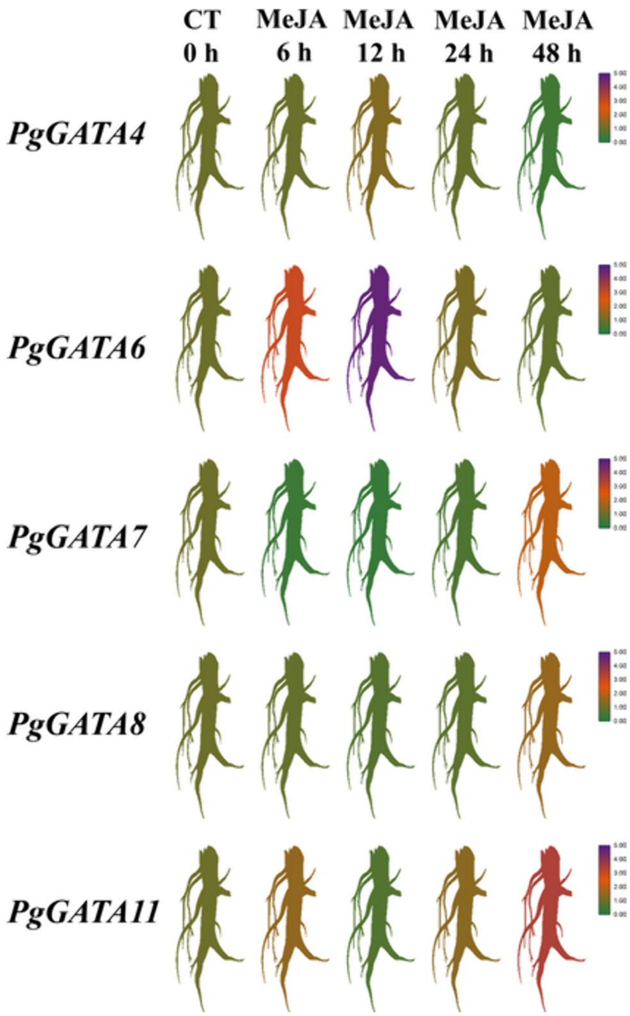


Fig. 7 Expression levels of five *PgGATAs* after MeJA treatment. MeJA treatment extended for 0, 6, 12, 24, and 48 from left to right. *GAPDH* was chosen as the reference gene to analyze the relative expression levels of *PgGATAs*

C were acidic except PgGATA14. Moreover, the number of amino acids and molecular weight of the proteins encoded by the members of this family were significantly different, which might directly impact their classification and function in evolutionary systems (Philippe et al. 2017; Zhu et al. 2022).

Domain analysis of genes revealed that each member contained only one GATA domain, which was in conformity with the corresponding structures in most plants, indicating that the structure of a *GATA* gene was relatively stable (Lai et al. 2022). The additional domains might play varying biological functions. The domain structure of groups A, B, and D is $CX_2CX_{18}CX_2C$, while that of group C is $CX_2CX_{20}CX_2C$, which was consistent with the previously reported domain structure and subfamily classification of the *GATA* gene family (Shi et al. 2022; Zhang et al. 2021; Zhu et al. 2020). The existence of these two zinc finger domains will directly affect the biological function of the *GATA* gene. In addition, our results were consistent with the previous studies that reported that CCT and TIFY were distributed only in group C. The CCT family genes are involved in the circadian clock, helping plants set suitable photoperiodic flowering pathways, and are related to biomass and grain output. Moreover, recent studies have shown that these genes are also associated with photosynthesis, nutrient use efficiency, and adversity resistance (Liu et al. 2020). The CCT domain, which regulates timing and flowering, and the TIFY domain, which regulates abiotic stress, are included in group C of the AtGATA family. This finding was consistent with the structure of group C members identified in the present study. A similar phenomenon was also observed in the pepper *GATA* gene family (Yu et al. 2021). H However, the CCT and TIFY domains were completely or partially missing in the GATA group C members in *Populus spp.* (Kim et al. 2021a) and *O. sativa* (Gupta et al. 2017). This finding might be attributed to evolution or selective shearing. PgGATA8 contains a unique ASXH domain, and the SmGATA03 gene containing the ASXH domain has also been identified in the studies on *Salix* genus (An et al. 2020).

The phylogenetic analysis of the evolution of *GATA* genes in *P. grandiflorum* showed that these genes could be divided into four groups similar to the *GATA* genes in most plant species (Du et al. 2022; Lai et al. 2022; Shi et al. 2022). However, in rice, the *GATA* gene family can be divided into seven subfamilies (Reyes et al. 2004). Therefore, our results were consistent with the conclusions of previous studies (An et al. 2020), that showed that groups V, VI, and VII disappeared after the differentiation of monocots and dicots. The evolution of species can directly influence the composition and classification of members of the GATA subfamily; however, there is a lack of direct evidence in this regard.

To comprehensively understand which proteins interact with the GATA family members, we found that 22 PgGATAs formed a protein–protein interaction regulatory network with corresponding 15 AtGATAs, indicating a similar interaction relationship in *P. grandiflorum* (Fig. 8). ZML1 protein was at the core of the whole protein–protein interaction network, highly similar to PgGATA9. The involvement of ZML1 and S-receptor kinase in the self-incompatibility (SI) genetic mechanism of *Brassica napus* highlighted its fundamental role (Duan et al. 2020). Shikata et al. (Shikata et al. 2004) found that ZML1 and ZML2 exhibit expression patterns similar to ZIM. Similarly, in the current study, we found that ZIM (PgGATA5) and ZML1 (PgGATA9) function identically or cooperatively. ZIM, ZML1 (At3G21175), and ZML2 (At1G51600) exhibit a close phylogenetic relationship. A previous study showed that the MYB/ZML protein complex can activate lignin genes and regulate lignin biosynthesis, especially

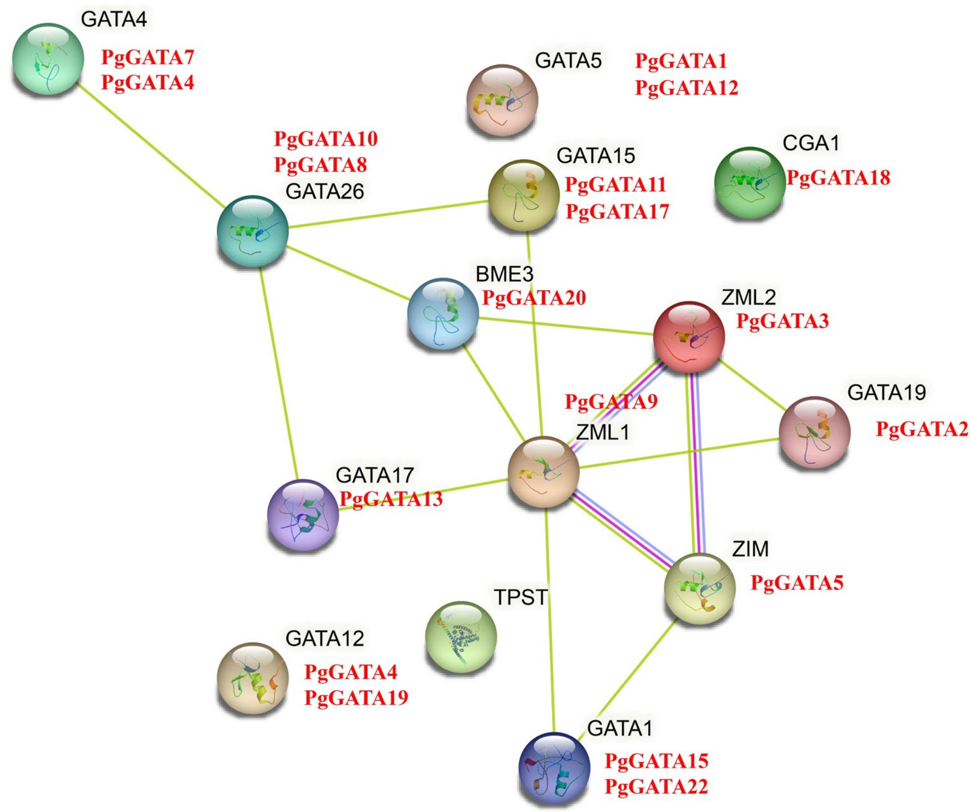


Fig. 8 Functional interaction network of PgGATAs in *P. grandiflorum* based on the orthologs in *Arabidopsis.* The light green, magenta, black, green, aqua, blue, and light sky blue lines represent the protein–protein associations predicted by text-mining, experimental determination, co-expression, gene neighborhood, gene co-occurrence, curated databases, and protein homology, respectively

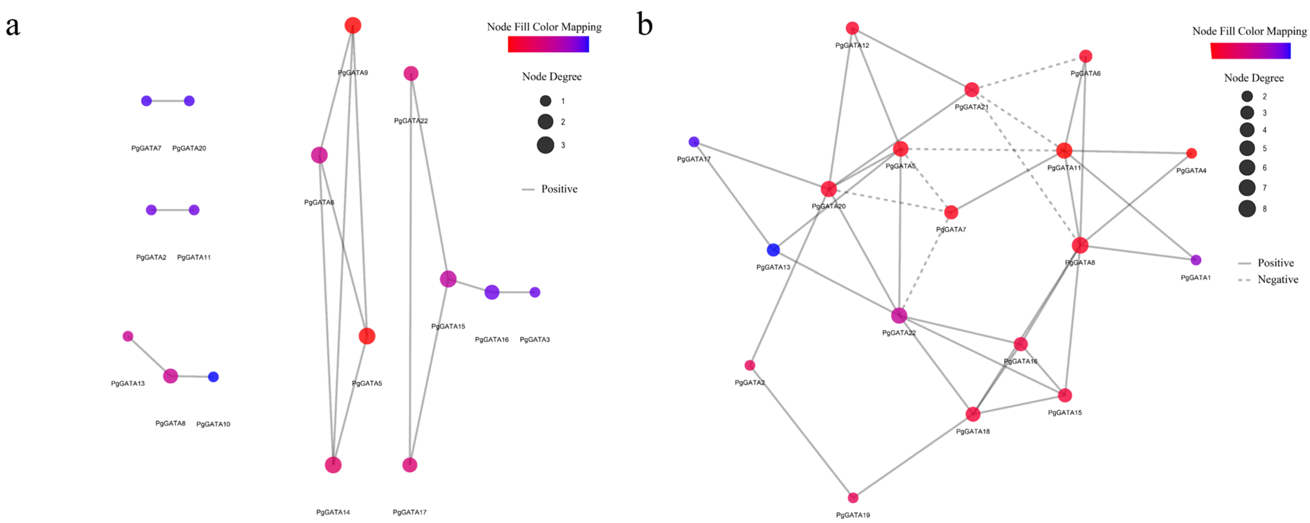


Fig. 9 The *PgGATA* co-expression network based on transcriptome data. **a** The *PgGATA* co-expression network in different tissues. **b** The *PgGATA* co-expression network in MeJA-treated roots at different time points. Shades of color represent the weights of the interacting genes, with red indicating a high weight and blue indicating a low weight of the associated genes. Solid lines indicate positive correlations and dashed lines indicate negative correlations

after MeJA treatment (Velez-Bermudez et al. 2015). The present study demonstrated that the *PgGATAs* are connected in a single network of co-expression, indicating that these genes maintain a functional correlation. This finding was in line with previous research that demonstrated a tendency of *GATA* genes to interact with each other (Peng et al. 2021). In *A. thaliana*, GATA TFs (GNC and CGA1) are direct and crucial targets for transcription downstream of ARF2. These factors play a key role in regulating processes such as greening, flowering time, and senescence

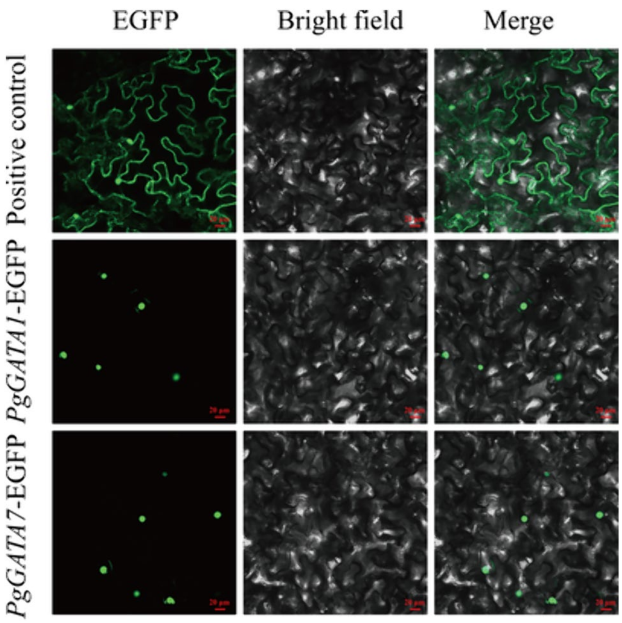


Fig. 10 Subcellular localization of PgGATA1 and PgGATA7. The green dots and the green curve correspond to the nucleus and cell membrane, respectively. Fluorescence signals of enhanced green fluorescent protein (EGFP) were detected in tobacco leaf epidermal cells. Left panel, EGFP image; middle panel, bright field; and right panel, merge image showing EGFP signal and bright field. Bar, 20 μm

(Richter et al. 2013). Consequently, *PgGATAs* might also interact with other TF proteins to govern plant growth and development.

MeJA can affect the levels of phenolic acids (Mubeen et al. 2022), flavonoids (Yan et al. 2022), saponins, and other substances (Kim et al. 2009, 2021b; Lei et al. 2022; Li et al. 2021; Chaudhary et al. 2015). Previous studies have shown the changes in the expression profiles of *GATA* gene family members after hormone treatment in pear (Manzoor et al. 2021) and chili (Yu et al. 2021). In addition, we analyzed the cis-acting elements in the current study and observed that most cis-acting elements were associated with MeJA treatment. Therefore, we analyzed the expression profiles of *GATA* genes in different tissue parts of MeJA-treated *P. grandiflorum*. The expression levels of *PgGATA6* peaked at 12 h after MeJA treatment and might be directly regulated by the MeJA signal. Furthermore, AtGATA23 (AT5G26930) and AtGATA29 (AT3G20750) contain degraded LLM domains that influence the formation of lateral roots by regulating auxin-corresponding modules (De Rybel et al. 2010). PgGATA11 is closely associated with AtGATA23 and AtGATA29 and contains a degraded LLM domain, which is highly expressed in the root at 48 h after MeJA treatment. Therefore, it is speculated that PgGATA11 is involved in the growth and development of *P. grandiflorum* roots.

Conclusion

The chromosomal location, duplication patterns, structural features, conserved domains and motifs, and expression levels of the *GATA* genes were assessed for genome-wide analyses in *P. grandiflorum*. We identified 22 *PgGATAs* and classified them into four groups based on phylogenetic analysis. Most PgGATAs are highly homologous to the identified AtGATAs. The transcriptome analysis, qRT-PCR, and gene co-expression network analysis showed that *PgGATA4*, *PgGATA6*, *PgGATA7*, *PgGATA8*, and *PgGATA11* were more responsive to MeJA-related signaling pathways. Overall, these results provided valuable information for further study of the structures, genetic evolutionary relationships, distinct patterns of gene expression, and MeJA treatment of GATA TFs in *P. grandiflorum.*

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00344-024-11468-8.

Author Contributions Weichao Ren: Conceptualization, Methodology, Formal analysis, Investigation, Writing—original draft. Lingyang Kong: Supervision, Investigation, Writing-review & editing. Shan Jiang, Lengleng Ma: Investigation, Validation. Honggang Wang, Xiangquan Li, Yunwei Liu: Methodology. Wei Ma: Conceptualization, Supervision, Project administration, Methodology, Writing-review & editing, Funding acquisition. Xueying Yan: Supervision, Investigation, Funding acquisition.

Funding This work was supported by the fund of National Key Research and Development Program of China (2021YFD1600902), Talent training project supported by the central government for the reform and development of local colleges and Universities (ZYRCB2021008), Heilongjiang Touyan Innovation Team Program (HLJTYTP2019001), Heilongjiang University of Traditional Chinese Medicine Outstanding innovative talents leading talent Project (2018RCL09), Heilongjiang Provincial research institute research business expenses project (CZKYF2022-F04).

Data Availability No data was used for the research described in the article.

Declarations

Competing Interest The authors confirm that there is no know conflict of interest associated with this publication.

References

An Y, Zhou YY, Han X, Shen C, Wang S, Liu C, Yin WL, Xia XL (2020) The GATA transcription factor GNC plays an important role in photosynthesis and growth in poplar. J Exp Bot 71(6):1969–1984. https://doi.org/10.1093/jxb/erz564

Bailey TL, Boden M, Buske FA, Frith M, Grant CE, Clementi L, Ren J, Li WW, Noble WS (2009) MEME SUITE: tools for motif discovery and searching. Nucleic Acids Res 37:W202–W208. https://doi.org/10.1093/nar/gkp335

Chaudhary S, Chikara SK, Sharma MC, Chaudhary A, Alam Syed B, Chaudhary PS, Mehta A, Patel M, Ghosh A, Iriti M (2015) Elicitation of diosgenin production in Trigonella foenum-graecum (Fenugreek) seedlings by methyl jasmonate. Int J Mol Sci 16(12):29889–29899. https://doi.org/10.3390/ijms161226208

Chen CJ, Chen H, Zhang Y, Thomas HR, Frank MH, He YH, Xia R (2020) TBtools: an Integrative toolkit developed for interactive analyses of big biological data. Mol Plant 13(8):1194–1202. https://doi.org/10.1016/j.molp.2020.06.009

Daniel-Vedele F, Caboche M (1993) A tobacco cDNA clone encoding a GATA-1 zinc finger protein homologous to regulators of nitrogen metabolism in fungi. Mol Gen Genet 240:365–373

Du K, Xia YF, Zhan DJ, Xu TT, Lu T, Yang J, Kang XY (2022) Genome-wide identification of the *Eucalyptus urophylla* GATA gene family and its diverse roles in chlorophyll biosynthesis. Int J Mol Sci 23(9):5251. https://doi.org/10.3390/ijms23095251

Duan ZQ, Zhang YT, Tu JX, Shen JX, Yi B, Fu TD, Dai C, Ma CZ (2020) The Brassica napus GATA transcription factor BnA5. ZML1 is a stigma compatibility factor. J Integr Plant Biol 62(8):1112–1131. https://doi.org/10.1111/jipb.12916

Feng X, Yu Q, Zeng JB, He XY, Liu WX (2022) Genome-wide identification and characterization of GATA family genes in wheat. BMC Plant Biol 22(1):372. https://doi.org/10.1186/s12870-022-03733-3

Guo J, Shi WP, Bai XH, Dai KL, Yuan XY, Guo PY, Hao CY (2021) Identification of GATA transcription factors in *Brachypodium distachyon* and functional characterization of BdGATA13 in drought tolerance and response to Gibberellins. Front Plant Sci 12:763665. https://doi.org/10.3389/fpls.2021.763665

Gupta P, Nutan KK, Singla-Pareek SL, Pareek A (2017) Abiotic stresses cause differential regulation of alternative splice forms of GATA transcription factor in rice. Front Plant Sci 8:1944. https://doi.org/10.3389/fpls.2017.01944

Kim OT, Bang KH, Kim YC, Hyun DY, Kim MY, Cha SW (2009) Upregulation of ginsenoside and gene expression related to triterpene biosynthesis in ginseng hairy root cultures elicited by methyl jasmonate. Plant Cell Tissue Organ Culture (PCTOC) 98(1):25–33. https://doi.org/10.1007/s11240-009-9535-9

Kim J, Kang S-H, Park S-G, Yang T-J, Lee Y, Kim OT, Chung O, Lee J, Choi J-P, Kwon S-J, Lee K, Ahn B-O, Lee DJ, Yoo S-i, Shin I-G, Um Y, Lee DY, Kim G-S, Hong CP, Bhak J, Kim C-K (2020) Whole-genome, transcriptome, and methylome analyses provide insights into the evolution of platycoside biosynthesis in *Platycodon grandiflorus*, a medicinal plant. Hortic Res. https://doi.org/10.1038/s41438-020-0329-x

Kim M, Xi H, Park S, Yun Y, Park J (2021a) Genome-wide comparative analyses of GATA transcription factors among seven Populus genomes. Sci Rep 11(1):16578. https://doi.org/10.1038/s41598-021-95940-5

Kim Y-K, Sathasivam R, Kim YB, Kim JK, Park SU (2021b) Transcriptomic analysis, cloning, characterization, and expression analysis of triterpene biosynthetic genes and triterpene accumulation in the hairy roots of Platycodon grandiflorum exposed to methyl jasmonate. ACS Omega 6(19):12820–12830. https://doi.org/10.1021/acsomega.1c01202

Lai DL, Yao X, Yan J, Gao AJ, Yang H, Xiang DB, Ruan JJ, Fan Y, Cheng JP (2022) Genomewide identification, phylogenetic and expression pattern analysis of GATA family genes in foxtail millet (Setaria italica). BMC Genomics 23(1):549. https://doi.org/10.1186/s12864-022-08786-0

Lei YY, Harris AJ, Wang AH, Zhao LY, Luo M, Li J, Chen HF (2022) Comparative transcriptomic analysis of genes in the triterpene saponin biosynthesis pathway in leaves and roots of Ardisia kteniophylla A. DC., a plant used in traditional Chinese medicine. Ecol Evolut 12(5):e8920. https://doi.org/10.1002/ece3.8920

Li L, Wang YF, Zhao MZ, Wang KY, Sun CY, Zhu L, Han YL, Chen P, Lei J, Wang Y, Zhang MP (2021) Integrative transcriptome analysis identifies new oxidosqualene cyclase genes involved in ginsenoside biosynthesis in Jilin ginseng. Genomics 113(4):2304–2316. https://doi.org/10.1016/j.ygeno.2021.05.023

Liu YL, Patra B, Pattanaik S, Wang Y, Yuan L (2019) GATA and phytochrome interacting factor transcription factors regulate light-induced vindoline biosynthesis in *Catharanthus roseus*. Plant Physiol 180(3):1336–1350. https://doi.org/10.1104/pp.19.00489

Liu HY, Zhou XC, Li QP, Wang L, Xing YZ (2020) CCT domain-containing genes in cereal crops: flowering time and beyond. Theor Appl Genet 133(5):1385–1396. https://doi.org/10.1007/s00122-020-03554-8

Lowry JA, Atchley WR (2000) Molecular evolution of the GATA family of transcription factors: conservation within the DNA-binding domain. J Mol Evol 50(2):103–115. https://doi.org/10.1007/s002399910012

Luo XM, Lin WH, Zhu SW, Zhu JY, Sun Y, Fan XY, Cheng ML, Hao YQ, Oh E, Tian MM, Liu LJ, Zhang M, Xie Q, Chong K, Wang ZY (2010) Integration of light- and brassinosteroid-signaling pathways by a GATA transcription factor in Arabidopsis. Dev Cell 19(6):872–883. https://doi.org/10.1016/j.devcel.2010.10.023

Ma CH, Gao ZJ, Hai MR, Chen JW, Zhang JJ, Zhang W, Shao JH, Yang SC, Zhang GH (2016) Candidate genes involved in the biosynthesis of Triterpenoid Saponins in Platycodon grandiflorum Identified by transcriptome analysis. Front Plant Sci 7:673. https://doi.org/10.3389/fpls.2016.00673

Manzoor MA, Sabir IA, Shah IH, Wang H, Yu Z, Rasool F, Mazhar MZ, Younas S, Abdullah M, Cai Y (2021) Comprehensive comparative analysis of the GATA transcription factors in four *Rosaceae* species and phytohormonal response in Chinese Pear (Pyrus bretschneideri) fruit. Int J Mol Sci 22(22):12492. https://doi.org/10.3390/ijms222212492

Mubeen B, Hasnain A, Mehboob R, Rasool R, Riaz A, Elaskary SA, Shah MM, Faridi TA, Ullah I (2022) Hydroponics and elicitation, a combined approach to enhance the production of designer secondary medicinal metabolites in *Silybum marianum*. Front Plant Sci 13:897795. https://doi.org/10.3389/fpls.2022.897795

Peng WY, Li W, Song N, Tang ZJ, Liu J, Wang YS, Pan S, Dai LY, Wang B (2021) Genome-wide characterization, evolution, and expression profile analysis of GATA transcription factors in *Brachypodium distachyon*. Int J Mol Sci 22(4):2026. https://doi.org/10.3390/ijms22042026

Philippe F, Pelloux J, Rayon C (2017) Plant pectin acetylesterase structure and function: new insights from bioinformatic analysis. BMC Genomics 18(1):456. https://doi.org/10.1186/s12864-017-3833-0

Rayko E, Maumus F, Maheswari U, Jabbari K, Bowler C (2010) Transcription factor families inferred from genome sequences of photosynthetic stramenopiles. New Phytol 188(1):52–66. https://doi.org/10.1111/j.1469-8137.2010.03371.x

Reyes JC, Muro-Pastor MI, Florencio FJ (2004) The GATA family of transcription factors in Arabidopsis and rice. Plant Physiol 134(4):1718–1732. https://doi.org/10.1104/pp.103.037788

Richter R, Behringer C, Zourelidou M, Schwechheimer C (2013) Convergence of auxin and gibberellin signaling on the regulation of the GATA transcription factors GNC and GNL in *Arabidopsis thaliana*. Proc Natl Acad Sci 110(32):13192–13197. https://doi.org/10.1073/pnas.1304250110

Romano O, Miccio A (2020) GATA factor transcriptional activity: Insights from genome-wide binding profiles. IUBMB Life 72(1):10–26. https://doi.org/10.1002/iub.2169

De Rybel B, Vassileva V, Parizot B, Demeulenaere M, Grunewald W, Audenaert D, Van Campenhout J, Overvoorde P, Jansen L, Vanneste S, Moller B, Wilson M, Holman T, Van Isterdael G, Brunoud G, Vuylsteke M, Vernoux T, De Veylder L, Inze D, Weijers D, Bennett MJ, Beeckman T (2010) A novel aux/IAA28 signaling cascade activates GATA23-dependent specification of

lateral root founder cell identity. Curr Biol 20(19):1697–1706. https://doi.org/10.1016/j.cub.2010.09.007
Scazzocchio C (2000) The fungal GATA factors. Curr Opin Microbiol 3(2):126–131. https://doi.org/10.1016/s1369-5274(00)00063-1
Shi M, Huang QK, Wang Y, Wang C, Zhu RY, Zhang SW, Kai GY (2022) Genome-wide survey of the GATA gene family in camptothecin-producing plant Ophiorrhiza pumila. BMC Genomics 23(1):256. https://doi.org/10.1186/s12864-022-08484-x
Shikata M, Matsuda Y, Ando K, Nishii A, Takemura M, Yokota A, Kohchi T (2004) Characterization of Arabidopsis ZIM, a member of a novel plant-specific GATA factor gene family. J Exp Bot 55(397):631–639. https://doi.org/10.1093/jxb/erh078
Subramanian B, Gao SH, Lercher MJ, Hu SN, Chen WH (2019) Evolview v3: a webserver for visualization, annotation, and management of phylogenetic trees. Nucleic Acids Res 47(W1):W270–W275. https://doi.org/10.1093/nar/gkz357
Szklarczyk D, Gable AL, Nastou KC, Lyon D, Kirsch R, Pyysalo S, Doncheva NT, Legeay M, Fang T, Bork P, Jensen LJ, von Mering C (2021) The STRING database in 2021: customizable protein–protein networks, and functional characterization of user-uploaded gene/measurement sets. Nucleic Acids Res 49(D1):D605–D612. https://doi.org/10.1093/nar/gkaa1074
Szklarczyk D, Kirsch R, Koutrouli M, Nastou K, Mehryary F, Hachilif R, Gable AL, Fang T, Doncheva Nadezhda T, Pyysalo S, Bork P, Jensen Lars J, von Mering C (2023) The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest. Nucleic Acids Res 51(D1):D638–D646. https://doi.org/10.1093/nar/gkac1000
Velez-Bermudez IC, Salazar-Henao JE, Fornale S, Lopez-Vidriero I, Franco-Zorrilla JM, Grotewold E, Gray J, Solano R, Schmidt W, Pages M, Riera M, Caparros-Ruiz D (2015) A MYB/ZML complex regulates wound-induced lignin genes in maize. Plant Cell 27(11):3245–3259. https://doi.org/10.1105/tpc.15.00545
Yan HQ, Zheng W, Wang Y, Wu YG, Yu J, Xia PG (2022) Integrative metabolome and transcriptome analysis reveals the regulatory network of flavonoid biosynthesis in response to MeJA in Camelliavietnamensis Huang. Int J Mol Sci 23(16):9370. https://doi.org/10.3390/ijms23169370
Yu CY, Li N, Yin YX, Wang F, Gao SH, Jiao CH, Yao MH (2021) Genome-wide identification and function characterization of GATA transcription factors during development and in response to abiotic stresses and hormone treatments in pepper. J Appl Genet 62(2):265–280. https://doi.org/10.1007/s13353-021-00618-3
Zhang CJ, Hou YQ, Hao QN, Chen HF, Chen LM, Yuan SL, Shan ZH, Zhang XJ, Yang ZL, Qiu DZ, Zhou XN, Huang WJ (2015a) Genome-wide survey of the soybean GATA transcription factor gene family and expression analysis under low nitrogen stress. PLoS ONE 10(4):e0125174. https://doi.org/10.1371/journal.pone.0125174
Zhang L, Wang YL, Yang DW, Zhang CH, Zhang N, Li MH, Liu YZ (2015b) Platycodon grandiflorus - an ethnopharmacological, phytochemical and pharmacological review. J Ethnopharmacol 164:147–161. https://doi.org/10.1016/j.jep.2015.01.052
Zhang Z, Ren C, Zou LM, Wang Y, Li SH, Liang ZC (2018) Characterization of the GATA gene family in Vitis vinifera genome-wide analysis expression profiles and involvement in light and phytohormone response. Genome 10(61):713–723. https://doi.org/10.1139/gen-2018-0042
Zhang KJ, Jia L, Yang DK, Hu YC, Njogu MK, Wang PQ, Lu XM, Yan CS (2021) Genome-wide identification, phylogenetic and expression pattern analysis of GATA family genes in cucumber (*Cucumis sativus* L.). Plants (Basel) 10(8):1626. https://doi.org/10.3390/plants10081626
Zhao T, Wu T, Pei T, Wang Z, Yang H, Jiang J, Zhang H, Chen X, Li J, Xu X (2021) Overexpression of SlGATA17 promotes drought tolerance in transgenic tomato plants by enhancing activation of the phenylpropanoid biosynthetic pathway. Front Plant Sci 12:634888. https://doi.org/10.3389/fpls.2021.634888
Zhu WZ, Guo YY, Chen YK, Wu DZ, Jiang LX (2020) Genome-wide identification, phylogenetic and expression pattern analysis of GATA family genes in *Brassica napus*. BMC Plant Biol 20(1):543. https://doi.org/10.1186/s12870-020-02752-2
Zhu XL, Wang BQ, Wang X, Wei XH (2022) Genome-wide identification, structural analysis and expression profiles of short internodes related sequence gene family in quinoa. Front Genet 13:961925. https://doi.org/10.3389/fgene.2022.961925

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